

1. Estimation by Random Sampling

Democratic politicians were astonished in 2010 when their early voter polls of sample voters showed Republican Scott Brown was favoured by a majority of voters and so would win the special election to fill the seat that late Democrat Teddy Kennedy had occupied for over 40 years. Based on their poll results, they mounted an intense, but ultimately unsuccessful, effort to save the seat for their party.

1.1. A Voter Poll

Suppose at some time before the election that p was the fraction of voters favouring Scott Brown. We want to estimate this unknown fraction p . Suppose we have some random process for selecting voters from registration lists that selects each voter with equal probability. We can define an indicator variable, K , by the rule that $K = 1$ if the random voter most prefers Brown, and $K = 0$ otherwise.

Now, to estimate p , we take a large number, n of random choices of voters. We define variables $K_1, K_2, \dots, K_n$ where K_i is the indicator variable for the event that the i th random voter chooses Brown. Since our choices are made independently, the K_i 's are independent. Now, let S_n be their sum.

$$S_n := \sum_{i=1}^n K_i \quad (1)$$

The variable $\frac{S_n}{n}$ describes this fraction. Most people, intuitively, and correctly, expect this sample to give a useful approximation of the unknown fraction, p .

So we will use the sample value $\frac{S_n}{n}$ as our *statistical estimate* of p . We know that S_n has a binomial distribution with parameters n and p ; we can choose n , but p is unknown.

How Large a Sample?

Suppose we want our estimate to be within 0.04 of the fraction, p , at least 95% of the time. This means we want

$$\Pr \left[\left| \frac{S_n}{n} - p \right| \leq 0.04 \right] \geq 0.95 \quad (2)$$

So we'd better determine the number, n , of times we must poll voters so that inequality (Equation 2) will hold. Chebyshev's Theorem offers a simple way to determine such an n .

S_n is binomially distributed. Equation (19.15), combined with the fact that $p(1-p)$ is maximized when $p = 1-p$, that is, when $p = \frac{1}{2}$, gives

$$\text{Var}[S_n] = n(p(1-p)) \leq n * \frac{1}{4} = \frac{n}{4} \quad (3)$$

Next, we bound the variance of $\frac{S_n}{n}$:

$$\begin{aligned} \text{Var} \left[\frac{S_n}{n} \right] &= \left(\frac{1}{n} \right)^2 \text{Var}[S_n] \quad (\text{Square Multiple Rule for Variance (19.9)}) \\ &\leq \left(\frac{1}{n} \right)^2 \frac{n}{4} \quad (\text{by 19.8}) \\ &= \frac{1}{4n} \end{aligned} \quad (4)$$

Using Chebyshev's bound and (Equation 4), we have:

$$\Pr \left[\left| \frac{S_n}{n} - p \right| \geq 0.04 \right] \leq \frac{\text{Var} \left[\frac{S_n}{n} \right]}{(0.04)^2} \leq \frac{1}{4n0.04^2} = \frac{156.25}{n} \quad (5)$$

To make our estimate with 95% confidence, we want the righthand side of (Equation 5) to be at most $\frac{1}{20}$. So we choose n so that

$$\frac{156.25}{n} \leq \frac{1}{20} \quad (6)$$

that is,

$$n > 3,125 \quad (7)$$

Section 19.6.2 describes how to get tighter estimates of the tails of binomial distributions that lead to a bound on n that is about four times smaller than the one above. But working through this example using only variance illustrates an approach to estimation that is applicable to arbitrary random variables, not just binomial variables.

1.2. Pairwise Independent Sampling

We can summarize the above reasoning into a slightly more general form with a basic result called the Pairwise Independent Sampling Theorem. In particular, we do not need to restrict ourselves to sums of zero-one valued variables, or to variables with the same distribution. For simplicity, we state the Theorem for pairwise independent variables with possibly different distributions but with the same mean and variance.

Theorem 19.4.1. (Pairwise Independent Sampling) *Let $G_1, \dots, G_n$ be pairwise independent variables with the same mean, μ , and deviation, σ . Define*

$$S_n := \sum_{i=1}^n G_i \quad (8)$$

Then

$$\Pr \left[\left| \frac{S_n}{n} - \mu \right| \geq x \right] \leq \frac{1}{n} \left(\frac{\sigma}{x} \right)^2 \quad (9)$$

Proof. We observe first that the expectation of $\frac{S_n}{n}$ is μ :

$$\begin{aligned} \mathbb{E} \left[\frac{S_n}{n} \right] &= \mathbb{E} \left[\frac{\sum_{i=1}^n G_i}{n} \right] \quad (\text{By the definition of } S_n) \\ &= \frac{\sum_{i=1}^n \mathbb{E}[G_i]}{n} \quad (\text{By linearity of expectation}) \\ &= \frac{\sum_{i=1}^n \mu}{n} \\ &= \frac{n\mu}{n} = \mu \end{aligned} \quad (10)$$

The second important property of $\frac{S_n}{n}$ is that its variance is the variance of G_i divided by n :

$$\begin{aligned}
\text{Var}\left[\frac{S_n}{n}\right] &= \left(\frac{1}{n}\right)^2 \text{Var}[S_n] \quad (\text{Square Multiple Rule for Variance (19.9)}) \\
&= \frac{1}{n^2} \text{Var}\left[\sum_{i=1}^n G_i\right] \quad (\text{definition of } S_n) \\
&= \frac{1}{n^2} \sum_{i=1}^n \text{Var}[G_i] \quad (\text{pairwise independent additivity}) \\
&= \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}
\end{aligned} \tag{11}$$

The Pairwise Independent Sampling Theorem provides a quantitative general statement about how the average of independent samples of a random variable approaches the mean. In particular, it proves what is known as the Law of Large Numbers: by choosing a large enough sample size, we can get arbitrarily accurate estimates of the mean with confidence arbitrarily close to 100%.

Corollary 19.4.2. *[Weak Law of Large Numbers] Let $G_1, \dots, G_n$ be pairwise independent variables with the same mean, μ , and the same finite deviation, and let*

$$S_n := \frac{\sum_{i=1}^n G_i}{n} \tag{12}$$

Then for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \Pr[|S_n - \mu| \leq \varepsilon] = 1. \tag{13}$$